

SpaSE-UNet3D: Sensor-Driven Wildfire Detection and Progression Prediction from VIIRS Multispectral Imagery

Nikolaos Mavros ¹ and Dimitrios Katsaros ^{1,*}

¹ Department of Electrical and Computer Engineering, University of Thessaly, Volos, Greece; nmavros@uth.gr (N.M.); dkatsar@uth.gr (D.K.)

* Correspondence: dkatsar@uth.gr

Abstract

Timely wildfire monitoring depends critically on optical and thermal infrared sensor observations from spaceborne instruments. The Visible Infrared Imaging Radiometer Suite (VIIRS) aboard Suomi-NPP and NOAA-20 is the primary sensor system underlying the TS-SatFire benchmark (2025), which consolidates multispectral VIIRS image stacks for three tasks: Active Fire (AF) detection, Burned Area (BA) mapping, and Fire Progression (FP) prediction. We make two contributions. First, a [label-quality audit across all tasks and splits](#) shows that many fires lack ground-truth annotations: 18 training fires and 2 test fires are excluded for AF, and the two unannotated test fires cannot be scored by any model. We additionally document the evaluation procedure, which differs from an independently implemented one in ways that make published and independently computed figures not directly comparable. We also document the BA label encoding in the released GeoTIFFs—a prerequisite for downstream modelling; [the BA task is audited only and no BA model is trained](#). Second, we propose SpaSE-UNet3D, a spatial Squeeze-and-Excitation 3D UNet using spatial-only (1,3,3) convolutions, which avoid temporal mixing in the convolutional path, and SE attention to reweight the VIIRS bands. Micro-averaged over all test pixels, it reaches $F1 = [F1 \pm SD]$ on the 15 annotated AF test fires and $[F1 \pm SD]$ on all 24 FP test fires, from two-day windows against the baselines' six, at 32.9 M and 24.3 M parameters on a single GPU. Code and results are publicly available.

Keywords: wildfire detection; VIIRS; active fire; fire progression; 3D U-Net; squeeze-and-excitation; remote sensing; deep learning; satellite imagery; label quality audit

1. Introduction

Wildfires are increasingly destructive across temperate and boreal biomes, and timely automated detection of active fire fronts, burned extents, and near-term fire progression is essential for operational response. Satellite-borne optical and thermal infrared sensors are the primary mechanisms enabling this monitoring at continental scale, with daily or twice-daily revisit and spectral bands specifically calibrated for fire radiometry. Among spaceborne systems, the Visible Infrared Imaging Radiometer Suite (VIIRS) aboard Suomi-NPP and NOAA-20 stands out: it provides 22 bands spanning the visible, near-infrared, and thermal infrared spectrum, of which five imaging-resolution bands at 375 m are directly exploited for active fire detection [1]. Barmountis et al. [2] survey the full landscape of optical sensor systems for early fire detection, distinguishing terrestrial, airborne, and spaceborne architectures and confirming that VIIRS and MODIS spaceborne sensors provide the most operationally scalable coverage for large-scale wildfire monitoring. [Alongside these](#)

Received:

Accepted:

Published:

Citation: . . . *Sensors* **2026**, *xx*, 0xx, 0.

Copyright: © 2026 by the authors.

Submitted to *Sensors* for possible open access publication under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

learned approaches, index-based methods applied to moderate-resolution optical sensors remain widely used for post-fire burned-area assessment: spectral indices such as the Normalized Burn Ratio have been applied to Landsat 8 imagery to delineate burning areas in protected forest [3], and comparable index-based procedures using Sentinel-2 data have been used to quantify burned extents in wildlife sanctuaries over multi-year periods [4]. These methods provide the operational baseline against which learned approaches are measured.

The recently released TS-SatFire dataset [5] consolidates several years of VIIRS observations into a common benchmark for three wildfire tasks: Active Fire (AF) pixel detection, Burned Area (BA) mapping, and Fire Progression (FP) prediction. Each fire event is represented as a temporal stack of multispectral VIIRS sensor images, with up to eight daytime reflectance and thermal channels plus complementary nighttime thermal bands — a design motivated by the demonstrated value of combining daytime optical sensor observations with nighttime thermal infrared data for round-the-clock fire tracking [6]. The dataset paper reports results for 12 baselines ranging from 2D U-Nets [7] to 3D transformer variants [8,9], using sensor temporal stacks of up to $TS = 6$ days. The strongest published AF baseline is SwinUNETR-3D at $TS = 2$ with $F1 = 0.823$; the strongest published FP baseline is U-Net-3D at $TS = 6$ with $F1 = 0.375$.

The benchmark has been used to rank eleven architectures against one another, but neither the completeness of its labels nor the evaluation procedure used to produce those rankings has been examined. Both, we find, materially affect the reported ordering. Our contribution is therefore twofold, and we state the ordering explicitly: the primary contribution is documentation of the benchmark’s evaluation protocol and label quality; the secondary contribution is an architecture, presented on efficiency grounds.

In this work we revisit TS-SatFire with two goals. The first is to examine the sensor data itself. A systematic audit of every fire in every split reveals a substantial number of fires with entirely missing labels across all three tasks, including two of the seventeen active-fire test fires. Fires that carry no annotation cannot be scored, so any test aggregate over the full split is determined for those fires by the convention adopted for empty ground truth rather than by model behaviour. We document the exclusion lists, the undocumented burned-area label encoding, and the effect of filtering at fire and day level, and we measure rather than assert the consequence for accuracy.

The second goal is architectural. The TS-SatFire sensor input is very short: operational constraints and paper conventions favour temporal windows of 1–2 days for AF and 2–6 days for FP. On such short windows, the implicit temporal prior that motivates full 3D convolutions in video understanding [10,11] or volumetric medical imaging [12] is **not obviously justified**: there is **little** inter-day sensor signal to decorrelate. Prior work in action recognition has shown that factorising 3D convolutions into separate spatial and temporal components yields better accuracy with fewer parameters [10,11], and **our design tests this intuition at its limit** by removing temporal mixing inside the residual blocks entirely. We propose SpaSE-UNet3D, a spatial Squeeze-and-Excitation 3D UNet that explicitly uses (1, 3, 3) convolutions in every residual block, so that the temporal dimension is preserved but never mixed across days. The block further uses SE [13] channel attention to reweight the VIIRS multispectral sensor bands per spatial context. Task-specific variants add an ASPP [14,15] bottleneck and a 27-channel auxiliary-feature input for fire progression.

We evaluate SpaSE-UNet3D on both the AF and FP tasks under the evaluation protocol defined in Section 6. On AF it reaches $F1 = [F1 \pm SD]$ at $TS = 2$ on the 15 annotated test fires, and on FP $F1 = [F1 \pm SD]$ at $TS = 2$ over the full 24-fire test set, in both cases using two-day observation windows where the published baselines use six, and training in under 12 hours on a single NVIDIA T4 accelerator. We do not present these figures as improvements over

the published baselines: as Section 6 establishes, the two sets of numbers are not computed under a common protocol. We treat the BA task as audit-only: the audit itself is a dataset contribution, but we do not present a learned BA model.

The objectives of this study are: (i) to quantify the completeness of the TS-SatFire labels across all three tasks and splits, and to document the burned-area encoding required to use them; (ii) to determine whether published baseline figures and independently computed figures measure the same quantity, and to isolate the contribution of each difference; (iii) to develop an architecture suited to the short observation windows the benchmark provides, and to characterise its computational cost as well as its accuracy; and (iv) to establish the run-to-run variability of both tasks, so that reported differences can be interpreted against measurement noise.

The rest of the paper is organised as follows. Section 2 reviews related work. Section 3 describes the TS-SatFire dataset. Section 4 presents our label-quality audit. Section 5 details the SpaSE-UNet3D architecture and its task-specific variants. Section 6 reports the experimental protocol and results. Section 7 discusses the findings and Section 8 concludes.

2. Related Work

Wildfire remote sensing with deep learning.

Early work on active fire detection from satellite imagery used per-pixel thresholds on thermal bands, typically VIIRS band I4 or the MODIS analogue [1,16]. Deep learning approaches in the last five years have moved to segmentation networks operating on multi-band VIIRS or Sentinel-2 stacks [17], and more recently to 3D and transformer architectures that incorporate temporal context [18]. Huot et al. [19] released the Next Day Wildfire Spread benchmark, with a related but coarser problem formulation than TS-SatFire. Across this literature, comparisons are made between architectures on a fixed pipeline, while the pipeline itself—label construction, cropping, and the aggregation of per-pixel counts into a reported score—is described briefly and varies between publications, so nominally identical metrics are not always comparable. We quantify that ambiguity on one benchmark in Section 6.2. Zhao and Ban [5] provide the most comprehensive comparison to date, benchmarking 12 architectures spanning U-Net [7], Attention U-Net [20], UNETR [9], SwinUNETR [8], and recurrent encoders on a unified TS-SatFire benchmark. Their reported figures are the reference point for this paper, and Section 6.2 examines the evaluation procedure that produced them.

Factorised spatiotemporal convolutions.

Tran et al. [10] and Qiu et al. [11] showed that factorising 3D convolutions into separate spatial $(1, k, k)$ and temporal $(k, 1, 1)$ stages outperforms full (k, k, k) kernels in video classification, with fewer parameters. SpaSE-UNet3D extends this principle to its extreme on short $(T \leq 2)$ windows by retaining only the spatial component. Our ablation (Section 6.5) supports the parameter-efficiency half of this claim; it does not resolve an accuracy difference.

Relation to existing 3D U-Net variants.

SpaSE-UNet3D shares its encoder–decoder skeleton with U-Net-3D [12], so it is worth stating where it departs from the 3D baselines it is evaluated alongside. U-Net-3D and Attention U-Net-3D apply isotropic $(3, 3, 3)$ convolutions throughout, and UNETR [9] and SwinUNETR [8] attend across the temporal axis at every stage. Both families were designed for volumetric medical data, where that axis is a spatial depth of many strongly correlated slices; on a stack of one or two daily acquisitions it is short and weakly correlated. We therefore remove learned mixing along that axis entirely, combining across days only

through global pooling in the SE modules and, for progression, a mean before the output head.

Attention in U-Nets.

Squeeze-and-Excitation [13] adds a lightweight channel-wise attention module that globally pools feature maps and produces per-channel gating weights. Roy et al. [21] extended this with concurrent spatial-channel SE for 2D semantic segmentation. Atrous Spatial Pyramid Pooling [14], later refined in DeepLabv3+ [15], aggregates multi-scale context via parallel dilated convolutions in the bottleneck. Both modules are standard in semantic segmentation, but their combination with spatial-only 3D convolutions for temporal-satellite tasks is, to our knowledge, new.

Data quality in remote-sensing benchmarks.

Prior work has documented label noise in land-cover benchmarks [22] and machine-learning test sets more broadly [23], showing that erroneous test labels can reorder benchmark rankings even when the underlying models are unchanged. Moreno-Ruiz et al. [24] document omission and commission errors in the MODIS burned-area products from which TS-SatFire's burned-area labels are partly derived, so label error in this benchmark is inherited as well as introduced. Wildfire benchmarks are also exposed to a form of incompleteness that is easy to miss: their ground truth is the output of another automated product rather than human annotation, so a fire that the source product failed to detect appears as a scene with no positive pixels, indistinguishable at the file level from a scene where nothing burned, and it trains without error. Our audit is in this tradition.

Sensor systems for wildfire monitoring in *Sensors*.

A body of work published in the MDPI *Sensors* journal is directly relevant to our setting. Barmpoutis et al. [2] provide a comprehensive survey of optical remote sensing systems—from ground-based infrared cameras to spaceborne VIIRS and MODIS sensors—for early fire detection, establishing the sensor taxonomy within which TS-SatFire sits. Chiang and Ulloa [6] demonstrate that combining the Sentinel-3 SLSTR optical/thermal instruments with the VIIRS day-night band (VIIRS-DNB) enables round-the-clock fire tracking, motivating TS-SatFire's inclusion of both daytime and nighttime VIIRS channels as model inputs. For burned area assessment, Moreno-Ruiz et al. [24] audit the omission and commission errors of MODIS-derived BA products, confirming that these remotely sensed labels carry systematic spatial and temporal biases that propagate to downstream benchmarks—a finding our own audit of TS-SatFire BA labels corroborates. On the deep learning side, Ghali et al. [25] benchmark CNNs and vision transformers for pixel-level wildfire segmentation from satellite imagery, and Casallas et al. [26] apply a 3D-CNN directly to satellite stacks for wildfire early alert, providing the closest architectural precedent to our SpaSE-UNet3D in the *Sensors* literature.

3. The TS-SatFire Dataset

TS-SatFire [5] consists of a set of fire events, each represented by a multi-day temporal stack of VIIRS observations. Per event, every day has: a VIIRS_Day GeoTIFF with eight spatial bands (six daytime reflectance and thermal bands plus two nighttime thermal bands), and a FirePred GeoTIFF with 19 auxiliary bands. The released rasters carry no band names, but the channel ordering is documented in the dataset publication [5], where the eight VIIRS bands and the nineteen auxiliary bands are enumerated as a single 27-channel sequence. We reproduce the correspondence in Table 1 because it is required to interpret any per-band analysis and is easy to lose when the two rasters are read separately. The

auxiliary bands fall into four groups: vegetation indices, current weather, terrain, and forecast weather.

Table 1. Correspondence between FirePred band index, the 27-channel sequence of the dataset publication [5], and the variable represented. Bands 1–8 of the 27-channel sequence are the VIIRS reflectance and thermal channels supplied in the separate VIIRS_Day raster.

FirePred band	Channel	Variable
1	9	NDVI
2	10	EVI
3	11	Total precipitation
4	12	Wind speed
5	13	Wind direction
6	14	Minimum temperature
7	15	Maximum temperature
8	16	Energy release component
9	17	Specific humidity
10	18	Slope
11	19	Aspect
12	20	Elevation
13	21	Palmer drought severity index
14	22	Land cover
15	23	Total precipitation, surface
16	24	Forecast wind speed
17	25	Forecast wind direction
18	26	Forecast temperature
19	27	Forecast specific humidity

Probing 62 fires, only the two precipitation bands (3 and 15) are ever uniformly zero—in 12 and 5 fires respectively—which is expected for events burning through periods without rainfall, and indicates valid measurements rather than missing data.

3.1. Labels

The VIIRS_Day GeoTIFF contains the label channels for all three tasks, encoded in dedicated bands of the same raster.

Active Fire labels.

Band 7 contains active-fire labels, encoded as VIIRS fire confidence levels [1]. Pixels with finite values ≥ 7 are active-fire pixels, which we threshold to a binary mask.

Burned Area labels.

Band 8 contains burned-area labels. Zhao et al. [5] describe how the BA labels are constructed — as the union of accumulated VIIRS active-fire detections and NIFC perimeters, with case-by-case manual quality control on the test set. The dataset paper does not document the precise storage convention used in the released GeoTIFFs, which we found necessary to verify before any modelling: in practice, band 8 encodes the burned mask implicitly, with finite values in the range [2,119, 908,660] corresponding to Julian-day codes or perimeter identifiers and NaN values corresponding to unburned pixels. The correct binary extraction is therefore

$$BA(p) = \mathbf{1}[\neg \text{NaN}(\text{band8}(p))] \quad (1)$$

and not a value threshold, where $\mathbf{1}[\cdot]$ is the indicator function and p ranges over the pixels of the raster.

Equation 1 is worth stating explicitly because the natural first attempt fails silently. Band 8 is a floating-point raster whose finite entries are large integers—dates or identifiers—and whose unburned pixels are NaN rather than zero. A reader who assumes the band holds a probability or a class score will apply a threshold such as $\text{band8}(p) > 0.5$; because every finite value exceeds any threshold in $[0, 1]$ and NaN comparisons evaluate to false, this happens to return the correct mask, and the error goes unnoticed until a different threshold or a different comparison is used. Conversely, converting NaN to zero before comparison—as `numpy.nan_to_num` does, and as the reference loader does elsewhere—turns unburned pixels into valid zeros and silently changes the class balance. The information carried by band 8 is therefore *presence*, not *magnitude*: a pixel is burned if and only if its entry is finite, and the numeric value encodes when or by which perimeter it burned, not how strongly. Every finite value we observed corresponds to a burned pixel; there are no finite non-burn entries. Any future burned-area or fire-progression work on TS-SatFire must use this encoding, and the derived progression target of Equation 2 inherits it.

Fire Progression labels.

Fire progression labels are not stored directly. They are derived from consecutive BA masks: the FP label for day T is the set of pixels burned on day $T + 1$ but not on day T ,

$$\text{FP}(T) = \text{BA}(T + 1) \setminus \text{BA}(T). \quad (2)$$

A window is usable only when both the last input day and the following day have valid (non-NaN) band 8.

3.2. Splits

The dataset paper defines the following splits.

AF / BA splits.

138 training fires, 13 validation fires, and 17 test fires (named historical events, e.g., *Dixie*, *Creek*, *Carr*). Two of the 17 test fires, *Calf Canyon* and *Mosquito*, are 2022 events.

FP splits.

The FP validation set extends the AF/BA list from 13 to 15 fires, adding 22713339 and 23301962; since 22713339 is present in the dataset, FP training has 137 fires rather than 138. 23301962 is absent from the Kaggle distribution, so the replicable FP validation set contains 14 fires. The test set is a separate collection of 24 US_2021 fires identified by geographic codes such as US_2021_CA3451712013120211011.

3.3. Spatial Coverage of the Centre Crop

TS-SatFire scenes are supplied at the full extent of each fire event, which in our copy of the dataset is 596×595 pixels for all but one event. As is standard practice in satellite segmentation pipelines, and as the reference implementation also does, we operate on a fixed 256×256 crop rather than the full scene, for memory reasons. A crop of this size retains approximately 19% of the imaged area. The reference implementation places the crop at a 30% offset from the scene origin; we centre it (Section 6.2).

This choice is not neutral, because fires are not centred within their scenes. We quantify the loss by counting, for every labelled positive pixel in each fire, whether it falls inside or outside the crop actually used. The results differ markedly between tasks. For active fire, a mean of 26.8% of labelled fire pixels fall outside the crop. For burned area and

fire progression, whose labels cover the accumulated extent of the event rather than the instantaneous front, the mean rises to 56.3%, and for several fires the crop retains no labelled positive pixels at all. The most severe active-fire case loses 86.2% of its labelled pixels; among the progression test fires, four lose more than 87% and one loses all of them.

The consequence is that for a substantial fraction of events the model is asked to predict a target that is largely absent from its input, and is scored on the discrepancy. This is a property of the evaluation setup rather than of any model, and it applies equally to every method evaluated under the same convention. Section 6.4 shows that the fraction of positive pixels lost to the crop predicts per-fire progression accuracy, and is a sufficient explanation for the low-scoring cases without appeal to fire behaviour. We report all results under the cropped protocol for consistency with the published baselines, and identify full-scene evaluation as necessary future work.

4. Label-Quality Audit

We audit every fire in every split of every task for label completeness. Missing labels occur at two distinct granularities, and conflating them discards usable data: some fires carry no annotation on any day of their temporal extent, while others are annotated on a subset of days only. We therefore audit and filter at both levels. For each fire we read the VIIRS_Day GeoTIFF for every available day and test whether the relevant label band contains any finite pixel. A fire with zero finite label pixels across its entire temporal extent carries no supervision and is excluded; a fire annotated on some days is retained, and individual unlabelled windows are rejected instead.

4.1. AF Audit

Table 2 summarises the audit for all three tasks.

Fire level. Of the 138 fires in the AF training pool, 18 have band 7 entirely NaN across every available day and are excluded; one validation fire (22712904) and two test fires fail the same test. The two test fires are *Calf Canyon* and *Mosquito*, both 2022 events for which active-fire ground truth was never included in the released data. They cannot be scored by any model, and any aggregate reported over the full 17-fire test set therefore depends for those fires on the convention adopted for empty ground truth rather than on model behaviour (Section 6.2). The audited splits contain 120 training, 12 validation and 15 test fires.

Day level. Excluding a fire outright because part of its temporal extent is unlabelled discards usable supervision. We instead retain partially annotated fires and reject individual windows: a window is admitted only if band 7 on its final day—the day to which the AF label refers—contains at least one finite pixel. Applied to the audited fire lists this rejects 143 training windows, leaving 1,719 training windows at TS = 2, each supervised by a verified label.

Without the fire-level exclusion, windows drawn from unannotated fires enter training with an all-negative target, since a NaN band read through the reference loader's `nan_to_num` conversion becomes a valid all-zero mask. The model is then supervised to suppress activations on scenes that do contain fire. Section 6.3 measures the size of this effect under our pipeline.

Sensitivity to the criterion. The fire-level criterion above is the weakest defensible one: a fire is removed only when it carries no supervision at all. A stricter alternative removes any fire whose labelled days fall below a fixed fraction of its temporal extent; at a 50% cut-off, three further fires are removed, leaving 117. We adopt the weaker criterion because day-level filtering already guarantees that every training window is verified, making fire-level removal of partially labelled fires unnecessary and wasteful of supervision.

4.2. BA Audit

Table 2 includes the BA audit. 49 of 138 training fires are excluded from BA training: 14 have no VIIRS_Day directory, 18 have band 8 entirely NaN across all days, and 17 have fewer than 50% of days with usable labels. All 5 excluded validation fires have band 8 entirely NaN. The AF and BA exclusion lists overlap substantially: 19 fires are excluded from both tasks, while 3 AF-excluded fires retain valid BA labels and 49 BA-excluded fires retain valid AF labels.

In addition to the missing-label audit, we also document the practical encoding of band 8 in the released GeoTIFFs. Zhao et al. [5] describe how the labels are produced (union of accumulated VIIRS active-fire detections and NIFC perimeters, with manual quality control on the test set) but do not specify the storage convention in the released rasters — a small but important detail for downstream users. Empirically we find that finite values in band 8 span the range [2,119, 908,660], consistent with Julian-day codes or perimeter identifiers from the MCD64A1 MODIS burned-area product rather than a binary or thresholded indicator. 100% of finite pixels are burned pixels; there are no finite non-burn values. The correct extraction of a binary BA mask is therefore presence-of-finite, as stated in Section 3. Documenting this encoding is a prerequisite for any future BA modelling on TS-SatFire.

4.3. FP Audit

Table 2 includes the FP audit. The FP audit is more subtle because FP labels are derived at the window level from consecutive BA masks. A fire is excluded if no usable window can be constructed, i.e., if it has zero consecutive (day T , day $T + 1$) pairs in which both days have valid BA. 48 training fires and 6 validation fires fail this test. [The submitted version of this paper additionally excluded eight fires from the test set, on a stated criterion of burned-area fraction below 0.001% inside the \$256 \times 256\$ crop. That criterion does not describe what was done: three of the eight exceed it, and the boundary is not monotone in the quantity it names. We therefore remove the test-set exclusion entirely and report all 24 test fires throughout. Six of the excluded fires do carry no usable window and contribute no counts under micro-averaging; the remaining two are scored normally.](#)

All exclusion lists are reported in full in the supplementary audit reports released with the repository.

Table 2. Label audit across all tasks and splits. *No dir.*: no VIIRS_Day directory present. *All NaN*: label band contains no finite pixel on any day. *<50% days*: label band finite on fewer than half the available days. The adopted fire-level criterion excludes the first two categories only; the third is reported for the sensitivity analysis of Section 4 and those fires are retained, with unlabelled windows rejected individually.

Task	Split	Total	No dir.	All NaN	<50% days	Excl.	Usable
AF	train	137	13	4	3	17	120
AF	val	14	1	1	0	2	12
AF	test	17	0	2	1	2	15
BA	train	137	13	18	17	31	106
BA	val	14	1	5	0	6	8
BA	test (AF)	17	0	7	1	7	10
BA	test (US_2021)	24	0	2	4	2	22
FP	train	137	—	—	—	48	89
FP	val	14	—	—	—	6	8
FP	test	24	—	—	—	6	18

Two conventions require comment. The AF and BA training pools are given as 137 rather than 138 because fire 22713339, listed in the AF/BA training split, is reassigned to fire-progression validation by the dataset paper; under the 138-fire convention used by our training pipeline the same audit excludes 18 fires and retains 120. Fire-progression exclusions are counted at the window level rather than the band level: a fire is unusable when no consecutive $(T, T+1)$ pair has valid burned-area labels on both days, so the three band-level columns do not apply. All exclusion lists are released in full with the code.

5. Method: SpaSE-UNet3D

SpaSE-UNet3D is a family of two closely related encoder–decoder architectures, one for AF detection and one for FP prediction. Both variants share four design principles: (i) spatial-only $(1, 3, 3)$ convolutions, (ii) SE channel attention [13] inside every residual block, (iii) a symmetric encoder/decoder with skip concatenations as in U-Net [7,12], and (iv) a combined Dice + Focal loss [27,28]. Task-specific variations handle the different input dimensionality, label density, and temporal-context budget of AF and FP. Fig. 1 shows the shared residual block side-by-side; Figs. 2 and 3 show the AF and FP variants respectively.

5.1. Shared Building Blocks

Spatial-only convolutions.

Every convolution in the encoder and decoder stages has kernel shape $(1, 3, 3)$ with padding $(0, 1, 1)$. The temporal dimension T is therefore preserved through all encoder/decoder stages and never mixed. For the short temporal windows used here ($T \in \{1, 2\}$ for AF, $T = 2$ for FP), temporal mixing adds parameters without providing signal; our internal ablations confirmed that full $(3, 3, 3)$ kernels hurt validation F1 on TS = 2.

SE channel attention.

After the second convolution in every residual block, features pass through a SEBlock3D module with reduction ratio $r = 8$. The module applies global average pooling over (T, H, W) , a two-layer MLP with dimensions $(\text{ch}, \text{ch}/r, \text{ch})$, and a sigmoid to produce per-channel gating weights which multiply the input. This lets the network learn which VIIRS spectral bands or auxiliary features are informative per spatial context.

Downsampling and upsampling.

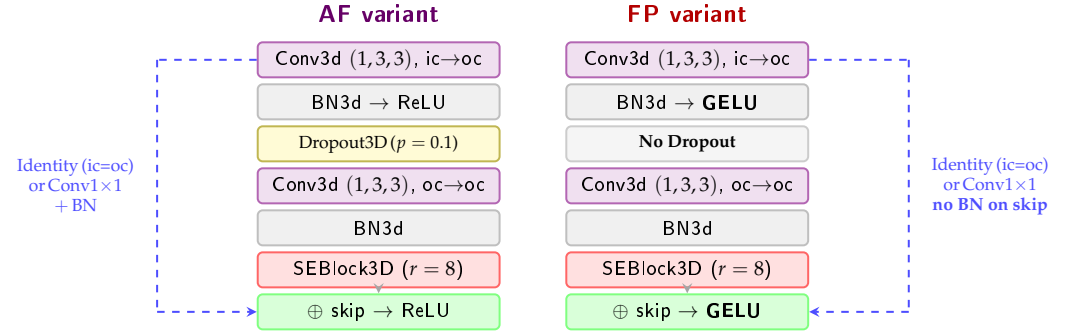
Encoder stages are separated by MaxPool3D(1, 2, 2), which halves the spatial dimensions but leaves T unchanged. Decoder stages upsample with ConvTranspose3D(1, 2, 2), and the upsampled features are concatenated with the matching-resolution encoder skip.

Residual blocks.

Each block applies two $(1, 3, 3)$ convolutions with batch normalisation [29] and a non-linear activation, followed by SE attention and a residual addition with the (possibly projected) input. Both variants of the ResBlock3D are shown in Fig. 1. The two variants differ only in the choice of activation (ReLU vs. GELU [30]) and the presence or absence of Dropout3D [31].

5.2. AF Variant

The AF variant takes an 8-channel input: the six daytime VIIRS bands and two nighttime VIIRS bands, stacked along the channel dimension of the temporal tensor. The encoder has four stages with output channels [64, 128, 256, 512]. The bottleneck is a single ResBlock3D that widens to 1024 channels. The decoder mirrors the encoder with channels



All convolutions use kernel (1, 3, 3) with padding (0, 1, 1). The temporal dimension T is preserved through the block but never mixed across days.

Figure 1. Internal structure of the ResBlock3D used in both SpaSE-UNet3D variants. Both blocks share spatial-only (1, 3, 3) convolutions and SE channel attention. The AF variant (left) uses ReLU activations and Dropout3D for regularisation; the FP variant (right) uses GELU and removes Dropout, which empirically hurt convergence on the smaller, sparser FP training set. The skip path either passes the input unchanged (when $ic = oc$) or applies a $1 \times 1 \times 1$ projection convolution.

[512, 256, 128, 64], each stage being a ResBlock3D operating on the concatenation of the upsampled feature map and the encoder skip. The full architecture is shown in Fig. 2.

ResBlock3D (AF).

The AF residual block (Fig. 1, left) applies two (1, 3, 3) convolutions with batch normalisation, ReLU activation, Dropout3D ($p = 0.1$) between them, an SE channel-attention module, and a residual addition with a projected (or identity) skip [32]. The Dropout3D inside the block regularises the network against the relatively small effective training set of 120 fires.

Deep supervision.

To force intermediate decoder representations to be spatially specific, the AF variant attaches an auxiliary segmentation head to the third decoder stage (d_3 , 256 channels), upsamples its output to the input resolution, and adds its loss to the total objective with weight $DS_WEIGHT = 0.3$. Both the main and auxiliary losses are computed on the last temporal frame only: the model produces predictions for all T input days, but the AF labels are only available for the final day, so we supervise only that frame.

Loss.

The AF training objective is a convex combination of Dice [27] and Focal loss [28]:

$$\mathcal{L}_{AF} = 0.5 \mathcal{L}_{Dice} + 0.5 \mathcal{L}_{Focal}(\alpha = 0.75, \gamma = 2) + 0.3 \mathcal{L}_{DS}. \quad (3)$$

The Dice term directly optimises the F1 surrogate; the Focal term handles the heavy class imbalance (fire pixels are typically $< 1\%$ of the patch).

5.3. FP Variant

The FP variant takes a 27-channel input: the 8 VIIRS bands, 18 FirePred auxiliary bands (band 3, total precipitation, is excluded), and 1 cumulative burned-area mask computed from all days up to T . The exclusion was made after observing that this band was identically zero in the fire used to probe the raster layout; the wider probe reported in Section 3 shows it to be zero in 12 of 62 fires and informative in the remainder, so restoring it is an obvious first step for future work. The encoder follows the same four-stage schedule [64, 128, 256, 512],

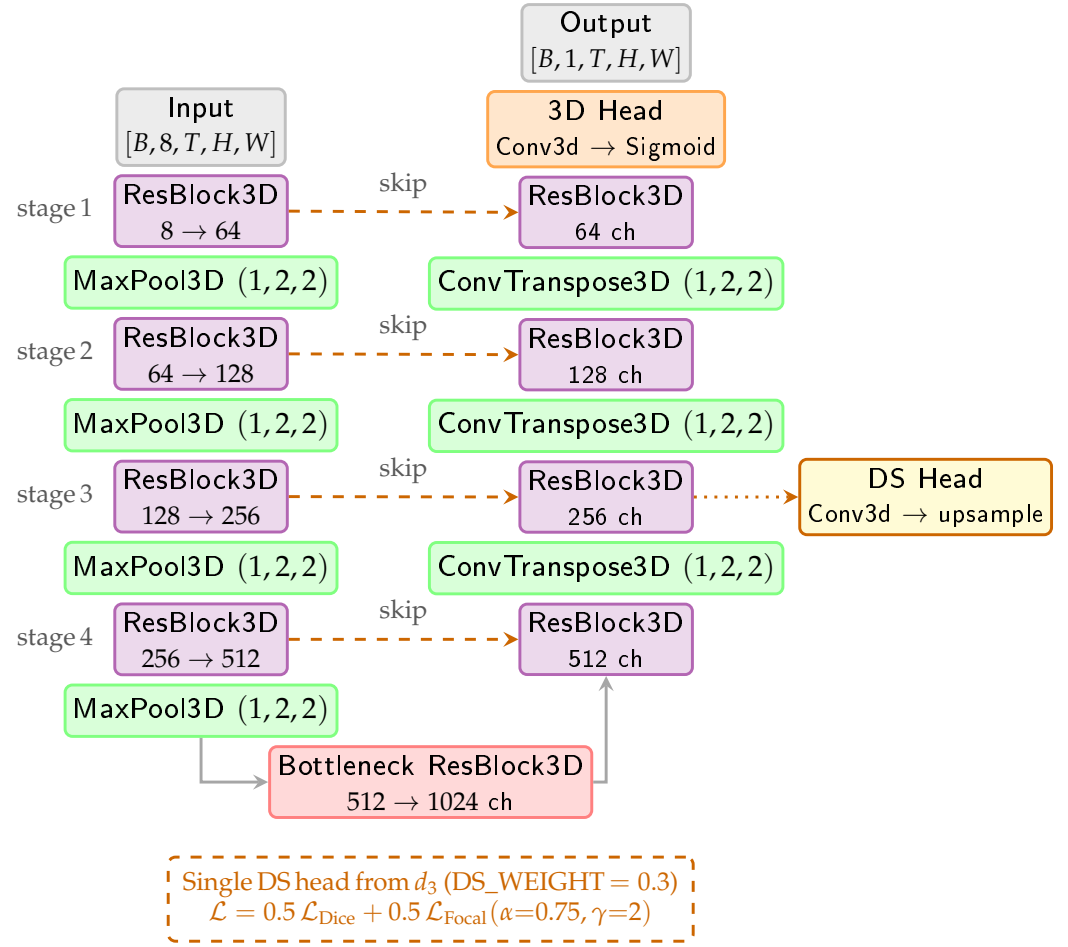


Figure 2. SpaSE-UNet3D AF variant. The encoder (left column) takes an 8-channel VIIRS input and downsamples spatially through four ResBlock3D stages with channels $[64, 128, 256, 512]$. A single ResBlock3D bottleneck widens to 1024 channels. The decoder (right column) mirrors the encoder with channel concatenation at every skip. A 3D head produces per-day fire masks. Deep supervision is applied at the third decoder stage.

but is preceded by a dedicated input-projection layer $\text{Conv3d}(27, 64, (1, 3, 3)) \rightarrow \text{BN} \rightarrow \text{GELU}$ that mixes the heterogeneous input channels before the first encoder block. The full architecture is shown in Fig. 3.

ResBlock3D (FP).

The FP residual block (Fig. 1, right) differs from the AF block in two ways. First, GELU [30] replaces ReLU as the activation; empirically we found that the sparser loss landscape of the FP objective benefits from the smoother gradient of GELU. Second, there is no Dropout3D [31] inside the block: on the much smaller effective FP training set (89 fires, with sparse window-level positives), Dropout hurt convergence. The skip connection also drops the BatchNorm on the projection to keep the residual path closer to identity.

ASPP3D bottleneck.

The FP variant replaces the plain bottleneck of the AF variant with a 3D Atrous Spatial Pyramid Pooling module [14,15]. Four parallel branches operate on the encoder output at stride 16:

- $\text{Conv3d}(512, 341, (1, 3, 3))$ with spatial dilation rate 1;
- the same with spatial dilation rate 6;

- the same with spatial dilation rate 12; 406
- global average pooling over (H, W) (preserving the temporal dimension T) followed 407
by Conv3d(512, 341, 1) and broadcast back to the spatial grid. 408

The four branch outputs (each 341 channels) are concatenated (for $4 \times 341 = 1364$ channels) 409
and fused with a Conv3d(1364, 1024, 1) $\rightarrow$ BN $\rightarrow$ GELU. The dilated branches capture 410
context at multiple spatial scales, which is critical because fire-front widths in the TS-SatFire 411
test set range from a few newly burned pixels to hundreds. 412

Final head. 413

The FP head first collapses the temporal dimension with a mean over T , then applies 414
a 2D head Conv2d(64, 32, 3) $\rightarrow$ BN2d $\rightarrow$ GELU $\rightarrow$ Conv2d(32, 1, 1) $\rightarrow$ Sigmoid. The 415
collapse is appropriate because the FP label itself is 2D (a single-day progression mask), 416
while the AF head remains 3D because AF labels are per-day. 417

No deep supervision. 418

We tested deep supervision for FP and removed it. The sparse, imbalanced FP labels 419
combined with the small effective training set made auxiliary losses introduce conflicting 420
gradients that hurt convergence. 421

Loss. 422

The FP training objective extends the AF loss with a weighted binary-cross-entropy 423
term: 424

$$\begin{aligned} \mathcal{L}_{\text{FP}} = & 0.5 \mathcal{L}_{\text{Dice}} + 0.3 \mathcal{L}_{\text{Focal}}(\alpha=0.85, \gamma=3) \\ & + 0.2 \mathcal{L}_{\text{BCE}}(w_+). \end{aligned} \quad (4)$$

The addition of BCE is necessary because the FP target is far sparser than AF: typical 425
windows have $< 0.05\%$ positive pixels. The positive weight w_+ is not a hand-tuned 426
constant: it is computed at preload time from the observed positive-pixel ratio of the 427
training split, and therefore varies slightly between runs (116.6–117.3 across our runs). This 428
removes one hyperparameter from the design and ties the term directly to the class balance 429
it is intended to correct. The higher Focal $\gamma = 3$ (vs. $\gamma = 2$ in AF) further downweights 430
easy negatives. 431

Test-time augmentation. 432

At inference, the FP variant uses $8\times$ test-time augmentation [33]: the input is passed 433
through horizontal flip, vertical flip, both, and three 90° rotations, plus the identity, and the 434
resulting logits are averaged before the sigmoid and threshold. 435

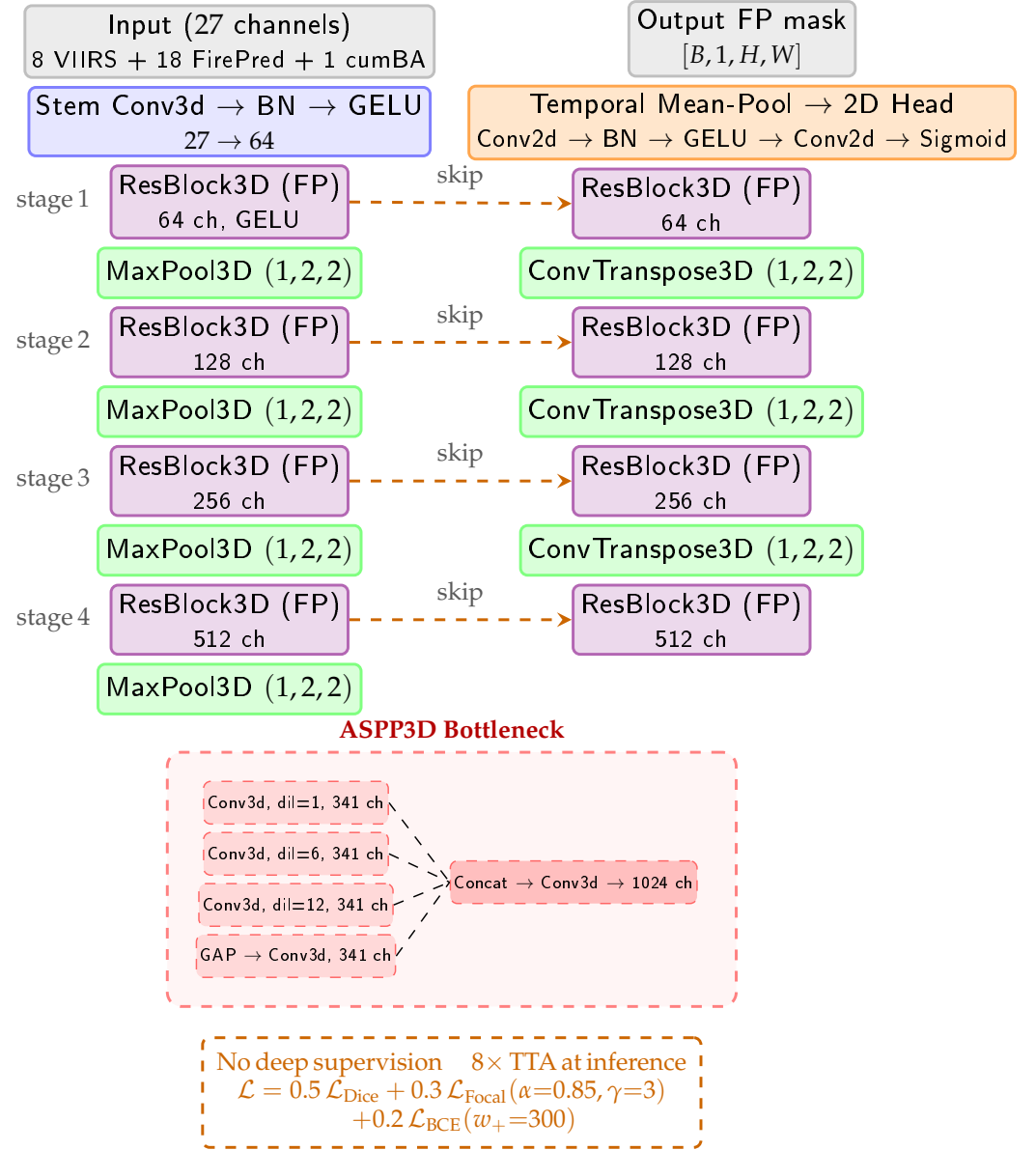


Figure 3. SpaSE-UNet3D FP variant. The encoder takes a 27-channel input combining VIIRS, FirePred auxiliary bands, and a cumulative burned-area mask. The bottleneck is replaced with an ASPP3D module that aggregates multi-scale context via parallel dilated convolutions and a global average pool. The final head collapses the temporal dimension via a mean-pool and applies a 2D convolutional head to produce a single-day FP mask. No deep supervision is used; 8× test-time augmentation is applied at inference.

6. Experiments

6.1. Implementation Details

All experiments are conducted on a single NVIDIA T4 GPU (16 GB VRAM) using PyTorch 2.10 with automatic mixed precision (AMP). A single training run takes between 3.5 and 6.5 hours depending on task and window length (Table 7); one run of each of the three configurations completes in approximately 14 hours.

Shared hyperparameters.

Batch size 8; patch size 256×256 (centre crop of the full scene); optimiser AdamW [34, 35] with learning rate 5×10^{-4} and weight decay 1×10^{-4} ; OneCycleLR [36] schedule with pct_start = 0.1 and cosine annealing; gradient clipping at norm 1.0. Each configuration

is trained from [N] (AF) and [N] (FP) independent random seeds; all reported figures are means over those runs with the standard deviation given alongside, and per-seed outputs are released with the code.

AF.

80 epochs for TS = 2, 100 epochs for TS = 1. VIIRS channels are normalised with per-band means and standard deviations computed once on a probe fire and reused across runs.

FP.

60 epochs. VIIRS channels use the same AF normalisation; FirePred auxiliary bands are normalised with per-band statistics computed at preload time on the training split and cached so that validation and test evaluation use identical statistics. Training data is augmented with horizontal and vertical flips plus 90° rotations; test-time augmentation is applied at inference as described in Section 5.

Sampling.

For AF we keep only training windows that contain at least 10 fire pixels and include at most 2 times as many negative windows as positive. For FP we use at least 3 burned pixels as the positive threshold and a negative-to-positive ratio of 1.5. These heuristics stabilise training without overfitting to the positive class.

Threshold selection.

For both tasks we sweep the inference threshold over the validation set and select the value that maximises validation F1, independently in each run, then apply it unmodified at test time. No test-set information enters the selection. The selected AF value varies widely between runs while test F1 does not, which we report as evidence of insensitivity in Section 6.3; the FP runs selected the same value throughout.

Metrics.

For all tasks we report pixel-level F1 and IoU (Jaccard index) aggregated over all pixels in all test fires (micro-average). Precision and recall at the selected operating point, and per-fire F1 and IoU, are also reported. Section 6.2 states the full evaluation protocol and its relationship to the reference implementation.

6.2. Evaluation Protocol

All baseline figures in this paper are quoted from Zhao et al. [5]; we did not retrain or re-evaluate them. Our own evaluation was implemented independently, and both works measure the same two tasks. The scoring procedures differ, however, and this section documents how, so that the figures we report are interpreted correctly.

The difference is visible in the reported metrics themselves. For metrics accumulated over a pixel population, F1 and the Jaccard index are related by a deterministic identity,

$$\text{IoU} = \frac{\text{F1}}{2 - \text{F1}}, \quad (5)$$

which our own measurements satisfy to four decimal places. The published pairs are related differently—SwinUNETR-3D at TS = 2 reports F1 = 0.823 with IoU = 0.727, where the identity would give 0.699—which indicates that the two works aggregate per-pixel counts differently: averaging a score computed per frame and per fire, rather than accumulating counts and computing the metric once. Both are legitimate choices, but they yield different

quantities, and figures computed under one should not be compared directly with figures computed under the other.

Reading the reference implementation identifies seven further differences (Table 3). Two matter beyond aggregation. Frames with empty ground truth are scored 1.0 rather than omitted, which affects the two test fires that carry no active-fire annotation (Section 4). And the two works define the positive class differently: $\text{nan_to_num}(\text{band } 7) > 0$ admits any non-zero confidence value, while we follow the VIIRS documentation [1] in placing the low-confidence boundary at 7.

Table 3. Differences between the TS-SatFire reference evaluation and the protocol used in this work.

Aspect	Reference implementation	This work
Aggregation	per-frame F1, averaged over frames, then fires	counts accumulated over all pixels, metric computed once
Empty ground truth	scored 1.0 ($\text{zero_division}=1.0$)	contributes no counts
AF label rule	$\text{nan_to_num}(\text{band } 7) > 0$	$\text{band } 7 \geq 7$
FP target	accumulated active fire, or its union with burned area, selected per fire by <code>label_sel</code>	$\text{clip}(\text{BA}(T+1) - \text{BA}(T), 0, 1)$
Spatial crop	30% offset	centred
NaN in band 8	counted as burned	treated as unburned
Threshold	fixed at 0.5	selected on validation
Test stride	stride = TS; all T frames scored	stride 1, final frame scored

Because these differences are separable, their cost can be measured. Table 4 applies them cumulatively to our AF model with the weights held fixed, so that only the measurement changes.

Table 4. Our AF model evaluated under progressively more of the reference protocol; model weights fixed throughout.

Configuration	Test fires	F1
This work (final frame, $\text{band } 7 \geq 7$)	15	0.7917
+ all frames scored	15	0.7714
+ reference label rule ($\text{band } 7 > 0$)	15	0.7567
+ the two unannotated test fires	17	0.6912
<i>SwinUNETR-3D, TS = 2, as published</i>	17	0.823

Scoring convention thus moves F1 by 0.10 on fixed weights, a range comparable to that separating the twelve published baselines (0.713–0.823). Our figures and the published ones are therefore not directly comparable, and we report ours under the protocol stated below without presenting them as improvements.

We evaluated our model under the reference protocol rather than retraining the baselines under ours, since a reimplementing would introduce our own training choices as confounds; retraining all baselines under a single protocol remains valuable future work.

All results below therefore use: counts accumulated over all pixels with the metric computed once; unannotated fires excluded and reported separately rather than scored; AF

labels as band 7 ≥ 7 on the final day of each window; FP targets as the clipped increment between consecutive burned-area masks; a centred 256×256 crop; and a threshold selected on validation and applied unmodified at test time. Per-seed outputs are released with the code.

6.3. Active Fire Results

Table 5 reports our test-set F1 and IoU at TS = 1 and TS = 2 alongside the twelve baselines of the dataset paper [5]. The two groups are listed separately because they are scored differently (Section 6.2): the baseline figures are reproduced as published, while ours are micro-averaged over the 15 annotated test fires at a validation-selected threshold. The table places the published landscape next to an independently measured result; it is not a ranking.

Under our protocol, SpaSE-UNet3D reaches $F1 = [F1 \pm SD]$ at TS = 2 over $[N]$ independent runs, and $F1 = [F1 \pm SD]$ at TS = 1. The difference between the two is small, and of the same order as the spread between repeated runs of the same configuration, so we treat the one-day and two-day results as substantially equivalent rather than ordering them. Per-fire inspection shows this equivalence is not uniform: individual fires shift by up to 0.022 in either direction between TS = 1 and TS = 2, with the largest losses on the weakest-signal events (Swedish, Blue Ridge). The second day therefore appears to add and remove information in roughly equal measure, which is consistent with there being little fire-specific signal in the additional frame.

Table 5. Active Fire detection results. Baseline figures are reproduced from [5] under the reference evaluation protocol; our figures are measured under the protocol of Section 6.2 on the 15 annotated test fires. The two groups are not directly comparable.

Model	TS	F1	IoU
<i>Published baselines, reference protocol, 17 test fires</i>			
U-Net (2D)	1	0.731	0.605
Att-U-Net (2D)	1	0.763	0.648
UNETR-2D	1	0.733	0.621
SwinUNETR-2D	1	0.774	0.660
GRU-3	6	0.713	0.601
LSTM-3	6	0.765	0.654
T4Fire	6	0.802	0.700
U-Net-3D	6	0.748	0.628
Att-U-Net-3D	6	0.770	0.654
UNETR-3D	6	0.811	0.706
SwinUNETR-3D	6	0.797	0.688
SwinUNETR-3D	2	0.823	0.727
<i>This work, protocol of Section 6.2, 15 annotated test fires</i>			
SpaSE-UNet3D	1	$[F1 \pm SD]$	$[IoU \pm SD]$
SpaSE-UNet3D	2	$[F1 \pm SD]$	$[IoU \pm SD]$

Threshold selection is performed independently in each run by sweeping the validation split and retaining the value that maximises validation F1, which is then applied unmodified at test time. The selected values differ substantially between runs while test F1 does not, which is itself the relevant result: performance is essentially flat in the threshold. A sweep over the test set confirms this, with F1 varying by less than 0.005 across the entire range $[0.20, 0.80]$ as precision and recall trade off symmetrically. In every run the

validation-selected threshold yielded test F1 within 0.0003 of the best achievable on the test set, so no advantage accrues from the selection procedure; we report the test sweep as evidence of robustness only, and it plays no part in selection.

Validation F1 rises above 0.80 within the first ten epochs and then oscillates as the operating point shifts between precision- and recall-favouring regimes, stabilising as the OneCycle schedule anneals the learning rate. Training and validation loss remain closely coupled throughout, with no divergence indicative of overfitting. The precision–recall panel is the empirical basis for the audit of Section 4: recall sits near 0.80 across training while precision oscillates widely, the asymmetry expected when some training windows are supervised by all-negative targets derived from unannotated scenes.

The aggregate test F1 is informative but masks per-fire variability. Fig. 4 breaks down test F1 across the 15 annotated test fires. 14 of 15 fires reach $F1 \geq 0.792$, and the median fire (*Double Creek*) reaches $F1 = 0.850$. Eight fires exceed $F1 = 0.85$. The single outlier is the low-intensity *Blue Ridge* fire at $F1 = 0.746$; manual inspection of its VIIRS scenes shows that the active-fire signal is faint, often confined to a single pixel, and consistently obscured by smoke plumes that drift from neighbouring confirmed-fire pixels.

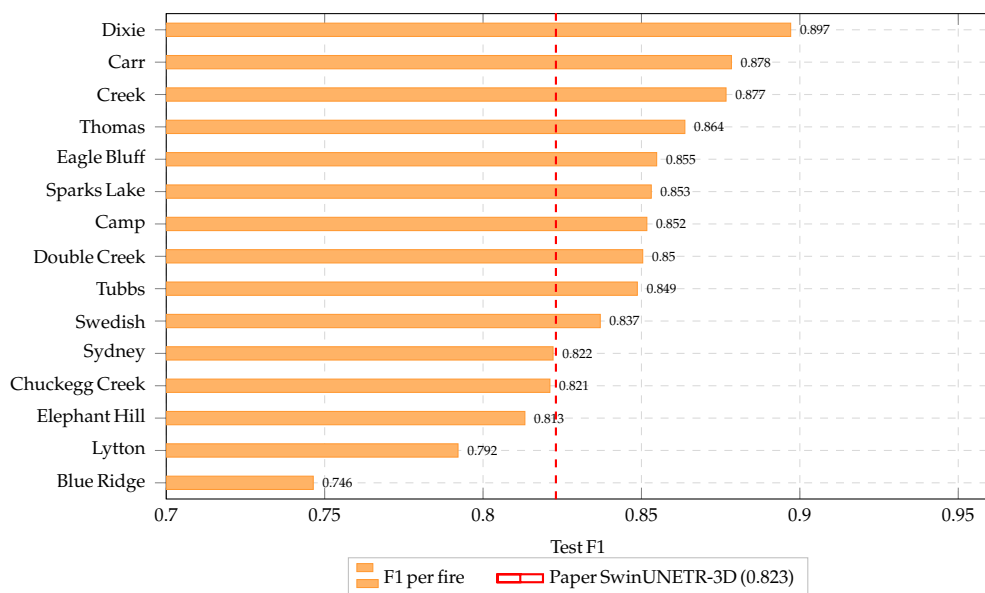


Figure 4. Per-fire test F1 for SpaSE-UNet3D at $TS = 2$ on the 15 annotated AF test fires, sorted by F1. Measured under the protocol of Section 6.2; no baseline reference line is shown, since published per-fire figures are not available and the aggregates are not computed under a common protocol.

To verify that the per-fire numbers reflect spatially coherent predictions and not artefacts of the aggregate F1, Fig. 5 shows true-positive (red), false-positive (green), and false-negative (blue) overlays for all 15 test fires at the optimal threshold. The dominant red colour across all panels confirms that the bulk of pixel-level predictions agree with the audited ground truth. False positives are typically small clusters along smoke or hot-soil boundaries (*Sydney*, *Chuckegg Creek*), and false negatives concentrate on the periphery of large active fronts (*Dixie*, *Creek*) where the radiometric signal is borderline. Crucially, no fire shows a systematic spatial bias, which is consistent with the symmetric precision/recall behaviour reported above.

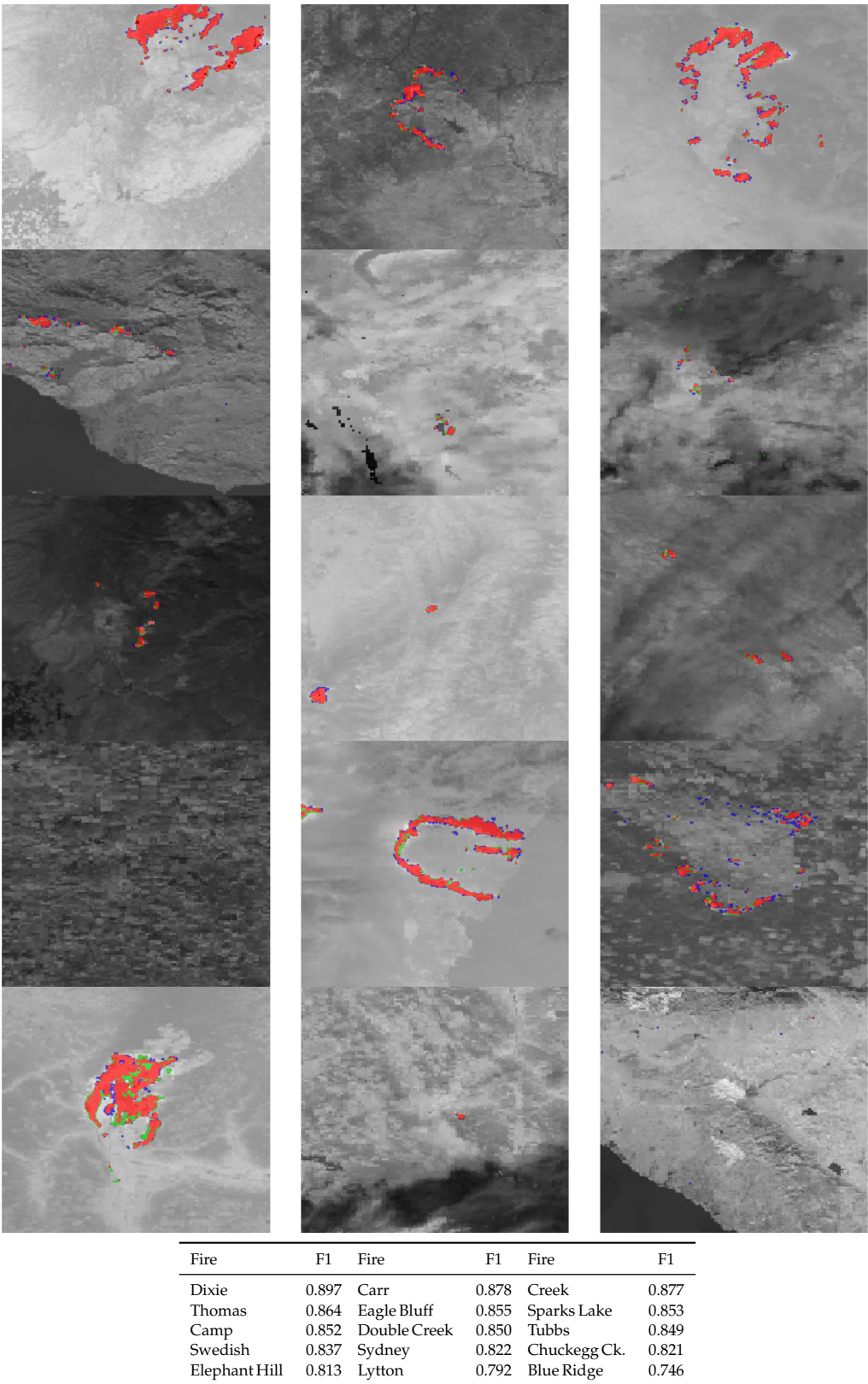


Figure 5. AF test-set predictions (TS = 2) for the 15 audited test fires, sorted by F1 from top-left to bottom-right. Each panel overlays the per-pixel confusion mask on the VIIRS day scene: red = true positive, green = false positive, blue = false negative. Per-fire F1 scores are listed in the table above.

Impact of the audit.

To isolate the effect of label filtering we trained two models differing in exactly one respect—whether windows drawn from unannotated days are retained with all-negative

558
559
560

targets—with all other settings held fixed. The audited model reaches test F1 = 0.8533 against 0.8514 for the unaudited one, a difference of 0.0019. This is an order of magnitude smaller than the figure reported in the submitted version, which compared models that also differed in training configuration; we correct it here. The explanation is that day-level filtering (Section 4) already rejects any window whose final day lacks a label, so only 71 of 1,790 windows differ between the two conditions. The practical implication is that day-level verification does most of the work, and that the failure mode appears at full magnitude only in a pipeline with neither filter. The value of the audit lies elsewhere: two of the seventeen AF test fires carry no annotation and cannot be scored, so any full-split aggregate depends for those fires on the empty-ground-truth convention rather than on the model.

6.4. Fire Progression Results

Following the reviewers' request, fire progression is now evaluated on the full official test set of 24 US_2021 fires rather than the 16-fire subset of the submitted version. The exclusion criterion stated there did not match the selection actually applied, so we remove it rather than restate it. Reporting all 24 fires costs 0.019 F1 relative to the subset.

Table 6 reports our result alongside the five published FP baselines. As in Section 6.2, the two groups are separated and are not directly comparable: the baselines are reproduced as published under the reference protocol, ours is measured under the protocol of Section 6.2.

Table 6. Fire Progression prediction results. Baseline figures are reproduced from [5] under the reference protocol; our figures are measured under the protocol of Section 6.2 over [N] independent runs. The two groups are not directly comparable.

Model	TS	F1	IoU
<i>Published baselines, reference protocol</i>			
U-Net-3D	6	0.375	0.338
SwinUNETR-3D	6	0.374	0.331
UNETR-3D	6	0.371	0.336
Att-U-Net-3D	6	0.354	0.312
SwinUNETR-3D	2	0.366	0.321
<i>This work, protocol of Section 6.2</i>			
SpaSE-UNet3D, all 24 test fires	2	[F1±SD]	
18 fires with a usable window	2	[F1±SD]	[IoU±SD]
16-fire subset of submitted version	2	[F1±SD]	[IoU±SD]

Run-to-run variance is worth reporting alongside the aggregate. Across [N] runs differing only in random seed, test F1 on the full test set has a standard deviation of [SD], against a range of 0.021 separating the five published baselines from one another. The seed noise of a single architecture is therefore of the same order as, or larger than, the entire spread between architectures, and single-run comparisons on this task cannot reliably distinguish them, our own included: the figure reported in the submitted version was one draw from this distribution.

Instability is visible during training as well. The epoch of best validation F1 varied by more than twenty epochs across runs while validation F1 at that epoch varied far less than test F1 did, indicating that validation performance is a weak predictor of test performance on this task. With only eight validation fires carrying usable progression windows (Section 4), this is expected, and it is a structural limitation of the benchmark rather than of any particular model. The contrast with active fire detection, where the same

architecture and pipeline give a far smaller spread, points to label sparsity and the small number of independent fire events—not the model—as the source.

The threshold was selected on the validation split in every run and applied unmodified at test time; the same value was selected in all runs. Validation F1 and IoU rise gently and monotonically over $[0.05, 0.75]$ with no sharp peak, so the choice is not critical.

Aggregate performance conceals wide per-fire variation (Fig. 6). The submitted version attributed the low-scoring cases to stochastic ember activity rather than coherent flame-front propagation; that explanation was speculative, and measurement identifies a concrete and sufficient cause instead: truncation of the fire by the 256×256 crop. Across the progression test set a mean of 56.3% of labelled positive pixels fall outside the crop (Section 3.3). Per-fire F1 correlates strongly and negatively with the fraction of positive pixels lost to the crop (Spearman $\rho = -0.685$, $p = 0.0034$). Fires scoring below 0.375 lose on average 66.7% of their positive pixels to the crop; those scoring above lose 34.6%. A model asked to predict progression on a crop containing a small fragment of the fire front is being evaluated on a task whose ground truth is largely absent from its input. The extreme case, *CA-4086* at $F1 = 0.007$, has eight usable windows in total with a progression increment occupying less than 0.01% of the crop; the model returns near-zero probability across the scene, which is the correct response to the input it is given, and is scored close to zero for it.

Centre-cropping is a standard convention adopted for memory reasons, but on this dataset it discards the majority of the progression signal, so per-fire scores partly reflect how much of each fire survives the crop. Full-scene evaluation is left to future work.

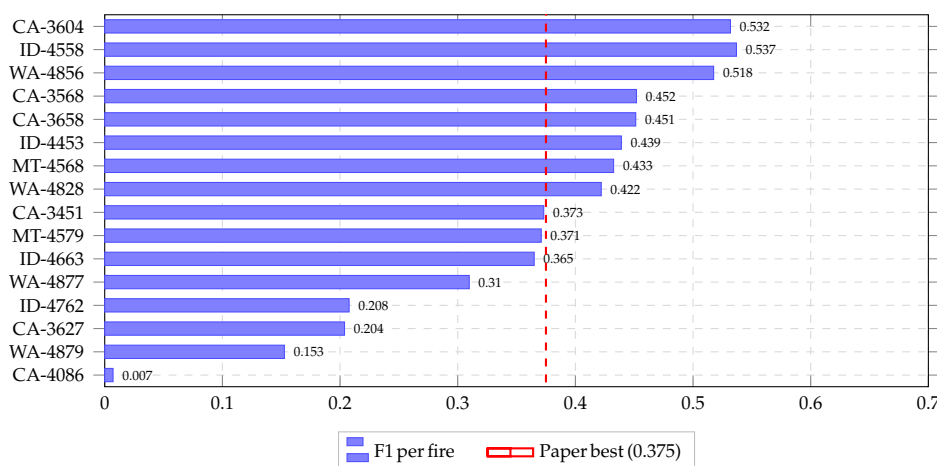
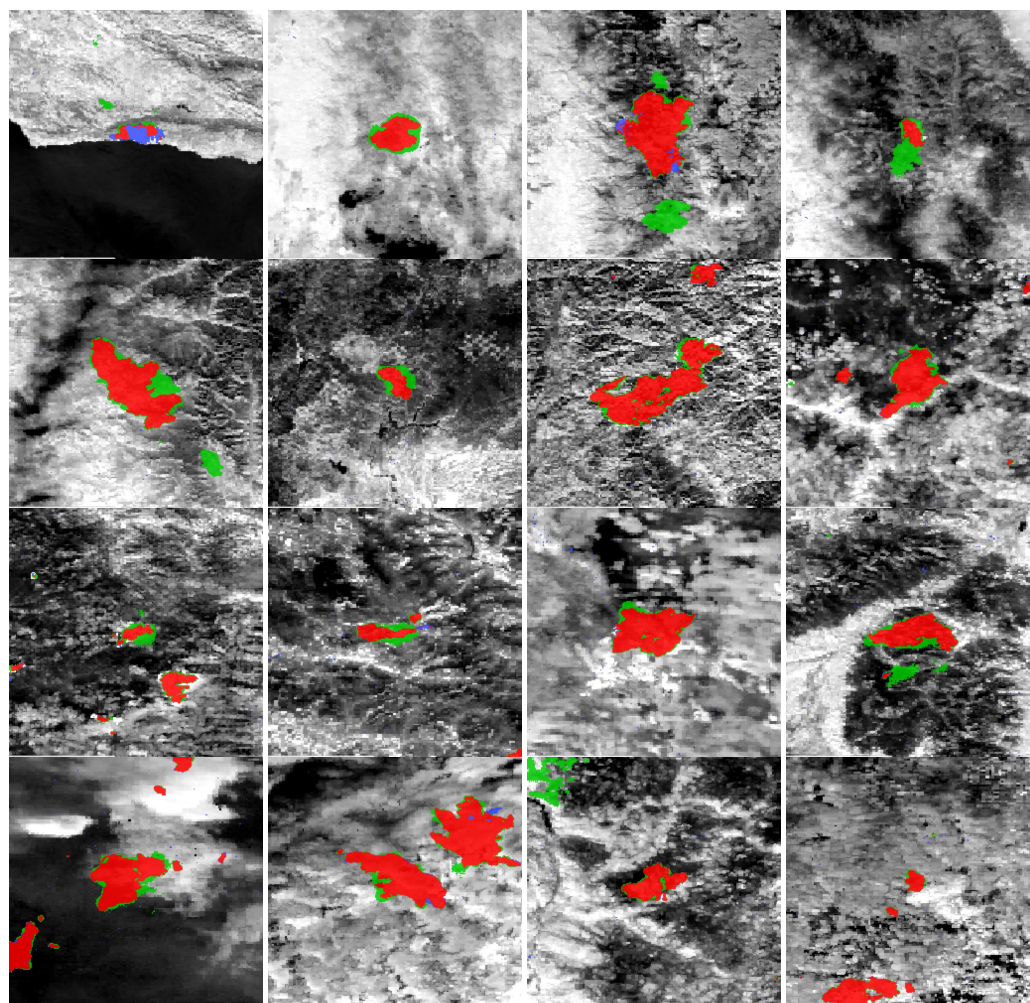


Figure 6. Per-fire test F1 for SpaSE-UNet3D on the US_2021 test fires, sorted by F1. Fires without a usable progression window contribute no counts and are omitted from the chart; the aggregate in Table 6 is over all 24 official test fires. No baseline reference line is shown, for the reasons given in Section 6.2.

To examine prediction quality qualitatively, Fig. 7 shows one representative window per fire for the test fires with a usable window, ordered by F1 from best (top-left) to worst (bottom-right). Most fires display predicted progression masks that closely follow the ground-truth perimeter shape: the model propagates the active-fire front in the direction implied by the cumulative burned-area channel and the FirePred terrain and weather covariates. Misalignment, where it occurs, manifests as a slight directional offset (*ID-4558*, *CA-3658*) rather than a complete shape mismatch. The lower-F1 fires reveal the dominant failure mode identified above: *ID-4762*, *CA-3627*, and *WA-4879* have very small ground-truth progression footprints relative to the crop, so that a single mispredicted patch dominates the score. These cases motivate reporting per-fire F1 alongside the aggregate, since a single truncated fire can shift the micro-averaged figure disproportionately.



Fire	F1	Fire	F1	Fire	F1	Fire	F1
CA-3604	0.532	ID-4558	0.537	WA-4856	0.518	CA-3568	0.452
CA-3658	0.451	ID-4453	0.439	MT-4568	0.433	WA-4828	0.422
CA-3451	0.373	MT-4579	0.371	ID-4663	0.365	WA-4877	0.310
ID-4762	0.208	CA-3627	0.204	WA-4879	0.153	CA-4086	0.007

Figure 7. FP prediction maps for the US2021 test fires with a usable progression window, sorted by F1 from top-left to bottom-right. Each panel overlays the predicted progression mask on the VIIRS day scene: **red** = true positive, **green** = false positive, **blue** = false negative. Per-fire F1 scores are listed in the table above.

Table 7 summarises the resource footprint for both tasks. SpaSE-UNet3D’s parameter count—24.28 M for FP and 32.88 M for AF—is in the same regime as the published 3D U-Net and transformer baselines rather than markedly smaller, and we make no claim of a parameter advantage over them. The efficiency argument rests instead on the temporal budget and the training cost: each configuration trains in between 3.5 and 6.5 hours on a single NVIDIA T4, using two-day observation windows where the published baselines use six.

Table 7. Training resource footprint on a single NVIDIA T4 GPU.

Task	Params	Epochs	Time (h)	Peak VRAM
AF (TS = 1)	32.88M	100	4.2	13.2 GB
AF (TS = 2)	32.88M	80	6.5	14.1 GB
FP (TS = 2)	24.28M	60	3.5	13.7 GB

6.5. Component Ablation

Four configurations were trained to isolate the contribution of the individual design choices: the full model, a variant replacing every spatial-only (1,3,3) kernel with a full (3,3,3) kernel, a variant with the squeeze-and-excitation modules removed, and a variant without the deep-supervision head. All arms used an identical 40-epoch budget and identical data, sampling, optimiser and schedule, and were evaluated on the validation split so that no test-set information entered any design decision. The full configuration was trained three times to estimate run-to-run variability; the three reduced configurations were trained once each.

Table 8. Component ablation, AF task, TS = 2, validation split, 40 epochs per arm.

Configuration	Val F1	Δ	Δ/σ	Params
Full model (3 runs)	0.8214 ± 0.0009	—	—	32.88 M
Full (3,3,3) convolutions	0.8205	−0.0010	1.1	89.36 M
Without SE attention	0.8219	+0.0005	0.6	32.44 M
Without deep supervision	0.8229	+0.0015	1.6	32.88 M

Run-to-run standard deviation on the full configuration is $\sigma = 0.0009$, and every reduced configuration falls within 1.6σ of it. At this sample size the experiment cannot separate the components, and we make no claim that any individual component improves accuracy; in particular, the submitted version’s assertion that full (3,3,3) convolutions reduce validation F1 is not supported and we withdraw it. Each reduced arm was trained once, so these are point estimates and a claim in either direction would require repeated runs per arm.

The comparison that is unambiguous concerns cost rather than accuracy. Replacing spatial-only kernels with full (3,3,3) kernels raises the parameter count from 32.88 M to 89.36 M, a factor of 2.72, while validation F1 remains within 1.1σ ; the SE modules account for 0.44 M parameters, or 1.3% of the model. On one- to two-day windows the additional temporal-mixing capacity is therefore not converted into performance, which supports the parameter-efficiency half of the finding reported in the action-recognition literature [10,11] rather than the accuracy half.

We did not ablate the fire-progression variant: the run-to-run standard deviation of [SD] on that task (Section 6.4) is far larger than any plausible single-component effect, so a single-run ablation would be uninterpretable. The FP design choices were made

sequentially on a validation split of eight usable fires without nested cross-validation, and a properly powered ablation—including of the cumulative burned-area input channel, which provides information some baselines may not receive—remains future work.

7. Discussion

The results above support three claims, stated with the strength the evidence allows.

Scoring convention affects reported figures as much as architecture does.

Applying the reference scoring procedure to fixed model weights moves F1 by 0.10 (Section 6.2), while the twelve published baselines span 0.11 between best and worst. Both works measure the same tasks, but through different aggregation and label conventions, and a difference of this size means figures from the two cannot be placed on a common axis. The identity $\text{IoU} = \text{F1} / (2 - \text{F1})$ provides a cheap check: a reported pair that does not satisfy it is not micro-averaged, and comparison with a micro-averaged pair is not meaningful.

Run-to-run variance on fire progression is large relative to the differences being compared.

Seeds differing in nothing else give a standard deviation of [SD] on the progression test set, against 0.021 separating the five published baselines. Single-run comparisons on this task therefore cannot reliably distinguish architectures, our own included. Active fire detection behaves differently under the same pipeline, with a spread two orders of magnitude smaller; the difference lies in label sparsity and in the small number of independent fire events, not in the models. Reporting seed variance before attributing a difference to an architecture seems to us the more transferable lesson of this study.

Spatial-only convolutions are inexpensive rather than demonstrably superior.

Our ablation (Section 6.5) does not resolve an accuracy difference between spatial-only and full 3D kernels at the sample size available, and we make no such claim. It does establish that full (3,3,3) kernels cost $2.72\times$ the parameters without measurable return on one- to two-day windows, and the near-equivalence of our TS = 1 and TS = 2 results points the same way. The architecture makes no assumption specific to VIIRS—it requires only a short stack of co-registered multispectral rasters and a per-pixel target—so it should transfer in principle to Sentinel-2, Sentinel-3 SLSTR or MODIS stacks, though we have not tested this.

Limitations.

Three limitations bear on how these results should be read. First, all conclusions rest on a single benchmark and a single sensor family. Second, the baselines were not retrained under our protocol, so the decomposition of Section 6.2 isolates the measurement effect on our own weights but does not establish how the baseline architectures would rank under a common evaluation. Third, the component ablation uses one run per reduced arm and is powered only to rule out large effects, and the fire-progression variant is not ablated at all; the seed variance measured on that task places a single-run ablation beyond useful resolution. We additionally train no burned-area model, so the question of whether the Julian-day codes in band 8 can support multi-class or regression targets is left open.

8. Conclusion

We presented **SpaSE-UNet3D**, a family of spatial-only 3D UNets with Squeeze-and-Excitation channel attention, together with the first systematic label-quality and evaluation-protocol audit of the TS-SatFire benchmark. Under the protocol defined in Section 6, SpaSE-UNet3D reaches $\text{F1} = [\text{F1} \pm \text{SD}]$ on active fire detection and $\text{F1} = [\text{F1} \pm \text{SD}]$ on fire

progression, using two-day observation windows where the published baselines use six, and training in under 15 hours on a single NVIDIA T4 GPU. We do not claim that these results exceed the published baselines: as Section 6 establishes, the two sets of figures are not computed under a common evaluation protocol, and we regard establishing that fact as the more useful contribution. The audit and protocol analysis are the paper's primary contribution: they establish that two of the seventeen active-fire test fires carry no annotation and cannot be scored, that the burned-area label encoding is undocumented in the dataset publication, and that evaluation convention moves F1 on this benchmark by more than the spread separating its published baselines. We release the exclusion lists, the protocol specification and the per-seed outputs so that future comparisons can be made on a stated and common basis.

Four directions follow directly from the limitations of this study. First, a properly powered ablation of the fire-progression variant, which the measured run-to-run standard deviation of [SD] places beyond the reach of single-run experiments. Second, retraining the baseline architectures under a single stated protocol, which would isolate architectural differences from measurement conventions. Third, full-scene rather than centre-cropped evaluation, since a mean of 56.3% of progression-labelled pixels fall outside the 256×256 crop and crop truncation predicts per-fire accuracy. Fourth, independent validation on other sensors and wildfire datasets, since all conclusions reported here rest on a single benchmark.

All code, checkpoints, and audit reports are publicly available at <https://github.com/mavroul1s/DeepFire-Forecaster>.

Author Contributions: Conceptualization, N.M.; methodology, N.M.; software, N.M.; validation, N.M.; formal analysis, N.M.; investigation, N.M.; writing—original draft preparation, N.M.; writing—review and editing, N.M. and D.K.; visualization, N.M.; supervision, D.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: Code, trained model checkpoints, and per-fire audit results are publicly available at <https://github.com/mavroul1s/DeepFire-Forecaster>.

Acknowledgments: The authors thank the TS-SatFire authors [5] for releasing their dataset and baselines in a form that makes this audit and comparison possible.

Conflicts of Interest: The authors declare no conflicts of interest.

References

- Schroeder, W.; Oliva, P.; Giglio, L.; Csiszar, I.A. The New VIIRS 375 m Active Fire Detection Data Product: Algorithm Description and Initial Assessment. *Remote Sensing of Environment* **2014**, *143*, 85–96.
- Barmapoutis, P.; Papaioannou, P.; Dimitropoulos, K.; Grammalidis, N. A Review on Early Forest Fire Detection Systems Using Optical Remote Sensing. *Sensors* **2020**, *20*, 6442. <https://doi.org/10.3390/s20226442>.
- Laosuwan, T.; Plybour, C.; Rotjanakusol, T. Inspection of Burning Areas Caused by Forest Fire in Doi Suthep-Pui National Park Through Data Obtained from Landsat 8 Satellite and Normalized Burn Ratio. *Agriculture and Forestry* **2023**, *69*, 81–95. <https://doi.org/10.17707/AgricultForest.69.3.06>.
- Uttaruk, Y.; Rotjanakusol, T.; Laosuwan, T. Burned Area Evaluation Method for Wildfires in Wildlife Sanctuaries Based on Data from Sentinel-2 Satellite. *Polish Journal of Environmental Studies* **2022**, *31*, 5875–5885. <https://doi.org/10.15244/pjoes/152835>.
- Zhao, Y.; Gerard, S.; Ban, Y. TS-SatFire: A multi-task satellite image time-series dataset for wildfire detection and prediction. *Scientific Data* **2025**, *12*, 1817. <https://doi.org/10.1038/s41597-025-06271-3>.

6. Chiang, S.H.; Ulloa, N.I. Mapping and Tracking Forest Burnt Areas in the Indio Maiz Biological Reserve Using Sentinel-3 SLSTR and VIIRS-DNB Imagery. *Sensors* **2019**, *19*, 5423. <https://doi.org/10.3390/s19245423>.
7. Ronneberger, O.; Fischer, P.; Brox, T. U-Net: Convolutional Networks for Biomedical Image Segmentation. In Proceedings of the Proc. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2015, pp. 234–241.
8. Hatamizadeh, A.; Nath, V.; Tang, Y.; Yang, D.; Roth, H.R.; Xu, D. Swin UNETR: Swin Transformers for Semantic Segmentation of Brain Tumors in MRI Images. In Proceedings of the Proc. MICCAI Brain Lesion Workshop (BrainLes 2021), LNCS vol. 12962, 2022, pp. 272–284.
9. Hatamizadeh, A.; Tang, Y.; Nath, V.; Yang, D.; Myronenko, A.; Landman, B.; Roth, H.R.; Xu, D. UNETR: Transformers for 3D Medical Image Segmentation. In Proceedings of the Proc. IEEE/CVF Winter Conf. Applications of Computer Vision (WACV), 2022, pp. 574–584.
10. Tran, D.; Wang, H.; Torresani, L.; Ray, J.; LeCun, Y.; Paluri, M. A Closer Look at Spatiotemporal Convolutions for Action Recognition. In Proceedings of the Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2018, pp. 6450–6459.
11. Qiu, Z.; Yao, T.; Mei, T. Learning Spatio-Temporal Representation with Pseudo-3D Residual Networks. In Proceedings of the Proc. IEEE Int. Conf. Computer Vision (ICCV), 2017, pp. 5534–5542.
12. Çiçek, Ö.; Abdulkadir, A.; Lienkamp, S.S.; Brox, T.; Ronneberger, O. 3D U-Net: Learning Dense Volumetric Segmentation from Sparse Annotation. In Proceedings of the Proc. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2016, pp. 424–432.
13. Hu, J.; Shen, L.; Albanie, S.; Sun, G.; Wu, E. Squeeze-and-Excitation Networks. In Proceedings of the Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2018, pp. 7132–7141.
14. Chen, L.C.; Papandreou, G.; Schroff, F.; Adam, H. Rethinking Atrous Convolution for Semantic Image Segmentation. *arXiv preprint arXiv:1706.05587* **2017**.
15. Chen, L.C.; Zhu, Y.; Papandreou, G.; Schroff, F.; Adam, H. Encoder-Decoder with Atrous Separable Convolution for Semantic Image Segmentation. In Proceedings of the Proc. European Conf. Computer Vision (ECCV), 2018, pp. 801–818.
16. Giglio, L.; Schroeder, W.; Justice, C.O. The Collection 6 MODIS Active Fire Detection Algorithm and Fire Products. *Remote Sensing of Environment* **2016**, *178*, 31–41.
17. Pinto, M.M.; Libonati, R.; Trigo, R.M.; Trigo, I.F.; DaCamara, C.C. A Deep Learning Approach for Mapping and Dating Burned Areas Using Temporal Sequences of Satellite Images. *ISPRS Journal of Photogrammetry and Remote Sensing* **2020**, *160*, 260–274.
18. Moskolai, W.R.; Abdou, W.; Dipanda, A.; Kolyang. Application of Deep Learning Architectures for Satellite Image Time Series Prediction: A Review. *Remote Sensing* **2021**, *13*, 4822.
19. Huot, F.; Hu, R.L.; Goyal, N.; Sankar, T.; Ihme, M.; Chen, Y.F. Next Day Wildfire Spread: A Machine Learning Data Set to Predict Wildfire Spreading from Remote-Sensing Data. *IEEE Trans. Geoscience and Remote Sensing* **2022**, *60*, 4412513.
20. Oktay, O.; Schlemper, J.; Le Folgoc, L.; Lee, M.; Heinrich, M.; Misawa, K.; Mori, K.; McDonagh, S.; Hammerla, N.Y.; Kainz, B.; et al. Attention U-Net: Learning Where to Look for the Pancreas. *arXiv preprint arXiv:1804.03999* **2018**.
21. Roy, A.G.; Navab, N.; Wachinger, C. Concurrent Spatial and Channel ‘Squeeze & Excitation’ in Fully Convolutional Networks. In Proceedings of the Proc. Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2018, pp. 421–429.
22. Pelletier, C.; Valero, S.; Inglada, J.; Champion, N.; Marais Sicre, C.; Dedieu, G. Effect of Training Class Label Noise on Classification Performances for Land Cover Mapping with Satellite Image Time Series. *Remote Sensing* **2017**, *9*, 173.
23. Northcutt, C.G.; Athalye, A.; Mueller, J. Pervasive Label Errors in Test Sets Destabilize Machine Learning Benchmarks. In Proceedings of the Proc. 35th Conf. Neural Information Processing Systems (NeurIPS) Datasets and Benchmarks Track, 2021.
24. Moreno-Ruiz, J.A.; García-Lázaro, J.R.; Arbelo, M.; Cantón-Garbín, M. MODIS Sensor Capability to Burned Area Mapping—Assessment of Performance and Improvements Provided by the Latest Standard Products in Boreal Regions. *Sensors* **2020**, *20*, 5423. <https://doi.org/10.3390/s20185423>.

25. Ghali, R.; Akhloufi, M.A.; Mseddi, W.S. Deep Learning and Transformer Approaches for UAV-Based Wildfire Detection and Segmentation. *Sensors* **2022**, *22*, 1977. <https://doi.org/10.3390/s22051977>.
26. Casallas, A.; Jiménez-Saenz, C.; Torres, V.; Quirama-Aguilar, M.; Lizcano, A.; Lopez-Barrera, E.A.; Ferro, C.; Celis, N.; Arenas, R. Design of a Forest Fire Early Alert System through a Deep 3D-CNN Structure and a WRF-CNN Bias Correction. *Sensors* **2022**, *22*, 8790. <https://doi.org/10.3390/s22228790>.
27. Milletari, F.; Navab, N.; Ahmadi, S.A. V-Net: Fully Convolutional Neural Networks for Volumetric Medical Image Segmentation. In Proceedings of the Proc. Int. Conf. 3D Vision (3DV), 2016, pp. 565–571.
28. Lin, T.Y.; Goyal, P.; Girshick, R.; He, K.; Dollár, P. Focal Loss for Dense Object Detection. In Proceedings of the Proc. IEEE Int. Conf. Computer Vision (ICCV), 2017, pp. 2999–3007.
29. Ioffe, S.; Szegedy, C. Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift. In Proceedings of the Proc. Int. Conf. Machine Learning (ICML), 2015, pp. 448–456.
30. Hendrycks, D.; Gimpel, K. Gaussian Error Linear Units (GELUs). *arXiv preprint arXiv:1606.08415* **2016**.
31. Srivastava, N.; Hinton, G.; Krizhevsky, A.; Sutskever, I.; Salakhutdinov, R. Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *Journal of Machine Learning Research* **2014**, *15*, 1929–1958.
32. He, K.; Zhang, X.; Ren, S.; Sun, J. Deep Residual Learning for Image Recognition. In Proceedings of the Proc. IEEE/CVF Conf. Computer Vision and Pattern Recognition (CVPR), 2016, pp. 770–778.
33. Wang, G.; Li, W.; Aertsen, M.; Deprest, J.; Ourselin, S.; Vercauteren, T. Aleatoric Uncertainty Estimation with Test-Time Augmentation for Medical Image Segmentation with Convolutional Neural Networks. *Neurocomputing* **2019**, *335*, 34–45.
34. Kingma, D.P.; Ba, J. Adam: A Method for Stochastic Optimization. In Proceedings of the Proc. Int. Conf. Learning Representations (ICLR), 2015.
35. Loshchilov, I.; Hutter, F. Decoupled Weight Decay Regularization. In Proceedings of the Proc. Int. Conf. Learning Representations (ICLR), 2019.
36. Smith, L.N.; Topin, N. Super-Convergence: Very Fast Training of Neural Networks Using Large Learning Rates. In Proceedings of the Proc. SPIE 11006, Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications, 2019, p. 1100612. <https://doi.org/10.1117/12.2520589>.